

A Further Note on the Statistical Meaning of Photon Number in a Thermal-Equilibrium Radiation System

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Abstract

This paper is a further supplement to the third part of the previous paper, concerning the statistical meaning of photon number in a thermal-equilibrium radiation system. It further points out that, from the perspective of structural ontology, the photon appears in statistical mechanics as a statistical count.

Preliminary Remarks

In the third part of the previous paper, on the statistical meaning of photon number in a thermal-radiation system [1], it was pointed out that the occupation number and the photon number are equivalent in statistical meaning, and that both belong to statistical quantities. On this basis, the present paper further explores the position of the photon in statistical physics from the perspective of structural ontology.

It is generally considered that the photon has wave-particle duality, and that it is grasped through physical pictures such as particle or wave. But within the framework of structural ontology, in what identity does the photon appear? This paper points out that, in structural ontology, the photon appears as a count in statistics.

Therefore, in structural ontology, the identity of the photon is no longer that of a particle or a wave, but that of a count in statistics. From this, we can quite naturally see that, under the description of a structural ontology, the core concern is the macroscopic statistical structure rather than the microscopic details of individuals.

Discussion

In statistical mechanics, if structural ontology is taken as the point of departure, then under this description, in what identity does the photon appear?

On the one hand, according to the previous paper, the photon number is manifested through the statistical quantity of occupation number. On the other hand, in the expression of the black-body radiation formula, it is not natural to imply the second Lagrange multiplier introduced by the constraint condition on the total photon number. Combining these two aspects, we may regard the photon as existing in statistics as a count.

From this point of view, several points may be drawn:

First, here the photon is no longer grasped as a physical picture such as a particle or a wave, but is even treated as a count in statistics. Here, there is no need for the photon to be regarded as something so concretely existing. Therefore, there is no need to use the picture of the original scene to describe the behavior of the photon. For example, there is no need to involve whether the total number of photons in a thermal-radiation system is conserved. Here it is easy to see that whether the photon number in an equilibrium system is conserved is not a question that needs to be judged, because under the description of structural ontology, the photon number itself is a statistical expression of the system's energy distribution.

Second, in statistical mechanics that takes structural ontology as its point of departure, the emphasis is placed on the importance of the overall structure of energy occupation rather than on the dynamical behavior of individuals. When the energy within a system changes, it is first manifested as an adjustment of the structure of energy occupation, and on the macroscopic level, this change is represented through a change in temperature.

In addition, structural ontology is not concerned with the detailed issues of individuals, such as interactions between photons or the current state of a single photon. It is concerned only with how energy is distributed over phase cells so that the entropy of the system reaches its maximum. Any macroscopic quantity is the result of statistics and does not involve the microscopic causal relations of individuals.

References

- [1] Bo Yang. A discussion of phase cells as the basic descriptive units of structure in statistical mechanics. 2025. DOI:10.5281/zenodo.18489919.